AG-SemiCD: A Semi-Supervised Remote Sensing Change Detection Framework Integrating Adaptive Confidence Filtering and Attention-Guided Feature Aggregation

Shenbo Liu, Dongxue Zhao, Ying Tan, and Lijun Tang, *Member, IEEE*

***Abstract*—Supervised change detection methods excel at extracting change features from remote sensing. However, their heavy reliance on large-scale, high-quality labeled data incurs high costs and time consumption, thereby restricting their practical application and scalability. To fully leverage limited labeled data and abundant unlabeled data, this article proposes AG-SemiCD, a semi-supervised change detection framework for remote sensing that combines adaptive confidence filtering with attention-guided feature aggregation. This framework incorporates an attention-guided feature aggregation network (AGNet) and a semi-supervised learning (SSL) strategy with adaptive confidence filtering. AGNet utilizes deep features to generate attention maps, guiding shallow feature learning to achieve more precise change feature extraction. The SSL strategy with adaptive confidence filtering effectively selects unlabeled samples with high-quality pseudo-labels for training. Simultaneously, an adaptive exponential moving average (A-EMA) mechanism is introduced to optimize the teacher model update process, maximizing the utilization of unlabeled data. Experimental results on three public datasets, namely LEVIR building change detection dataset (LEVIR-CD), WHU building dataset (WHU-CD), and multiscale remote sensing change detection dataset (MSRS-CD), demonstrate that AG-SemiCD significantly outperforms eight state-of-the-art (SOTA) methods across multiple key evaluation metrics, including $F1$ score and intersection over union (IoU), showcasing robust performance advantages and broad application potential.**

***Index Terms*—Attention mechanism, change detection, remote sensing, semi-supervised learning (SSL).**

I. Introduction

REMOTE sensing change detection (RSCD) is a core technique in the field of Earth observation, aiming to accurately identify and extract dynamic land-cover changes by quantitatively analyzing bitemporal or multitemporal images of the same geographical area acquired at different times [1]. This technology plays a crucial role in a wide range of applications, including urban expansion monitoring [2], forest resource management [3], disaster damage assessment [4], and agricultural planning [5].

In recent years, with the rapid advancement of deep learning, data-driven models represented by convolutional neural networks (CNNs) [6], [7], [8] and vision transformers (ViTs) [9], [10], [11] have significantly propelled progress in RSCD. Due to their strong nonlinear representation capability and end-to-end learning paradigm, these models have substantially overcome the performance bottlenecks of traditional approaches. Specifically, CNNs leverage local receptive fields and hierarchical feature extraction mechanisms, excelling at capturing fine-grained textures and edge details. In contrast, ViTs effectively model long-range global dependencies through self-attention mechanisms, demonstrating great potential in understanding large-scale and complex semantic changes. However, the success of these methods is highly contingent upon large-scale, high-quality, and pixel-level aligned change detection datasets. In practical applications, remote sensing image interpretation not only demands extensive professional expertise but is also time-consuming and labor-intensive, resulting in high annotation costs and difficulties in scaling datasets. The contradiction between abundant data and scarce labels significantly constrains the widespread deployment of supervised learning models in actual RSCD operations.

To address this issue, semi-supervised learning (SSL) has gradually emerged as an important research direction in RSCD. SSL aims to improve model performance by exploiting large amounts of unlabeled data while relying on only a small set of labeled samples, thus significantly reducing annotation costs. Existing semi-supervised learning change detection (SSL-CD) methods can generally be categorized into three groups: generation-based methods, pseudo-label-based methods, and consistency regularization-based methods. Generation-based approaches typically characterize change regions through reconstruction errors, latent space discrepancies, or adversarial generative training. S4GAN adopts a dual-branch network architecture that integrates multitask learning and self-training mechanisms, and mitigates feature artifacts caused by label scarcity through adversarial loss [12]. Souly et al. [13] employ GANs to generate a large amount of synthetic visual data, forcing real samples to cluster in

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The authors are with the School of Physics and Electronic Science, Changsha University of Science and Technology, Changsha 410114, China (e-mail: liusb@csust.edu.cn; zhao-dx@stu.csust.edu.cn; tanyinger@stu.csust.edu.cn; tanglj@csust.edu.cn).

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feature space and thereby enhancing multiclass pixel classification capability. Pseudo-label-based methods iteratively enhance model performance by treating high-confidence predictions on unlabeled data as additional supervisory signals. The AdaSemiCD algorithm proposed by Ran et al. [14] introduces quantifiable pseudo-label quality evaluation metrics and incorporates category reweighting and ambiguous-region enhancement mechanisms, significantly improving the reliability of pseudo-labels. In addition, the RCL method strengthens feature extraction by designing contrastive loss tailored to change regions and utilizes uncertainty analysis on unlabeled data to filter high-quality pseudo-labels, effectively improving change detection accuracy [15]. Consistency regularization-based methods impose constraints on model outputs under different views by applying geometric transformations or feature perturbations to unlabeled samples, thereby encouraging the learning of intrinsic data structures. Han combines multiscale fusion and attention mechanisms to progressively refine feature representations, and introduces a mean teacher framework that stabilizes student model updates using an exponential moving average (EMA) strategy [16]. STCRNet proposed by Wang et al. [17], built upon VGG16 as the backbone, collaboratively integrates self-training mechanisms and consistency regularization constraints to achieve efficient change detection.

Although existing SSL-CD methods offer certain advantages in scenarios with scarce samples, they still face two core challenges when dealing with complex remote sensing scenes. First, most methods directly employ general-purpose backbone networks, failing to fully mine the structural supervision embedded in sparse labels. Consequently, model performance remains limited in scenarios involving complex backgrounds, blurred boundaries, and multiscale changes. Second, in semi-supervised training, the absence of effective quality control mechanisms can lead to unstable or erroneous pseudo-label predictions, introducing noise. This noise accumulation and confirmation bias subsequently degrade model performance.

To solve the aforementioned problems, this article proposes a semi-supervised change detection framework for remote sensing imagery named AG-SemiCD, which integrates adaptive confidence filtering and attention-guided feature aggregation. This framework aims to maximize the utilization of limited labeled data and abundant unlabeled data to achieve efficient feature extraction, robust pseudo-label learning, and high-precision change identification. In terms of network architecture, an attention-guided feature aggregation network (AGNet) is designed. This network integrates a dual-stream attention-aware fusion module, a high-semantic attention map generator (AMG), and an attention-guided optimization module, constructing a deep-to-shallow feature aggregation mechanism. By utilizing attention maps generated from deep features to explicitly guide the refinement of shallow features, AGNet is capable of precisely capturing multiscale change features even in the absence of extensive supervisory signals. In terms of the semi-supervised training strategy, a strategy based on adaptive pseudo-label filtering and adaptive exponential moving average (A-EMA) is proposed. This strategy adopts the mean teacher framework, employing AGNet as both the teacher and student models. The student model learns basic features from limited labeled data, while the teacher model is dynamically updated via A-EMA to mitigate the impact of noise from the student network and maintain stable supervisory signals. Furthermore, an adaptive confidence pseudo-label filter is designed, which combines entropy-based uncertainty with change ratio constraints to perform quality control on pseudo-labels, reducing the impact of pseudo-label noise on model training at the source. The main contributions of this article are as follows.

1) A semi-supervised change detection framework, AG-SemiCD, is proposed to jointly leverage limited labeled data and large-scale unlabeled data, improving change detection performance while reducing annotation costs.
2) An AGNet is designed, which enhances change feature extraction under scarce-label conditions through deep-to-shallow attention guidance, enabling high-quality, multi-scale, and robust change representation learning.
3) A semi-supervised training strategy combining adaptive confidence filtering and adaptive EMA is introduced to effectively suppress pseudo-label noise propagation, alleviate confirmation bias, and improve model stability and training efficiency.
4) Extensive experiments on three public datasets—LEVIR building change detection dataset (LEVIR-CD), WHU building dataset (WHU-CD), and MSRS-CD—demonstrate that AG-SemiCD significantly outperforms eight state-of-the-art (SOTA) methods in key metrics such as $F1$ score and intersection over union (IoU), validating its superiority and versatility.

The remainder of this article is organized as follows. Section II briefly introduces related work on supervised RSCD and SSL-CD methods. Section III details the design of the AG-SemiCD network. Section IV presents a comprehensive experimental analysis of AG-SemiCD. Section VI concludes the article.

II. Related Work

A. Supervised RSCD

With the breakthroughs of deep learning in visual feature representation, deep learning-based RSCD methods have developed rapidly. Early research primarily focused on introducing CNNs to RSCD tasks and verifying their superior feature extraction capabilities. Caye Daudt et al. [8] proposed three fully convolutional Siamese structures: FC-EF, FC-Siam-conc, and FC-Siam-diff, laying the foundation for the application of deep learning in change detection. To further enhance feature discriminability, attention mechanisms have been widely incorporated. Chen et al. [18] proposed DASNet, which embeds dual spatial and channel attention modules within a Siamese network, significantly reinforcing the expression of change features in multitemporal high-resolution imagery. In addition, Chen et al. [19] introduced FCCDN, which improves boundary delineation of change regions by incorporating constraint mechanisms and self-supervised strategies during feature extraction and fusion.

Despite their strong performance, CNN-based methods are limited by their restricted receptive fields, making it dif-

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ficult to capture long-range dependencies. To address this issue, transformer-based architectures have been explored. Zhang et al. [20] proposed SwinSUNet, a Siamese U-shaped network based on the Swin transformer, to improve the modeling of global spatiotemporal dependencies. Chang et al. [21] introduced MDIPNet, which alleviates the high computational cost and insufficient semantic utilization of Transformers in remote sensing imagery through a multiscale dual spatial interaction mechanism. However, pure transformer models often incur high computational overhead and tend to be difficult to optimize on small-scale datasets. To balance local detail modeling and global semantic understanding, hybrid CNN–transformer architectures have emerged. Chen et al. [10] proposed the BiT model, which uses CNNs for initial feature extraction and transformer encoders to enhance spatiotemporal contextual modeling. Jiang et al. [22] further integrated graph neural networks to improve the modeling of structured change patterns.

With the maturation of vision foundation models, methods based on large-scale pretrained models have also garnered attention. Ding et al. [23] proposed SAM-based change detection (SAMCD) based on FastSAM, transferring the generalization capability of large models to RSCD scenarios, thereby significantly enhancing adaptability to complex remote sensing scenes. Li et al. [24] utilized contrastive language-image pre-training (CLIP) as a frozen knowledge base, achieving efficient fusion of foundation model knowledge and RSCD features through bitemporal adapter branches and connection modules.

Overall, although supervised learning methods have achieved significant breakthroughs in RSCD, their performance relies heavily on large-scale, high-quality pixel-level annotations. Annotating bitemporal remote sensing imagery is not only time-consuming and labor-intensive but also requires domain expertise. This severely restricts the promotion and large-scale application of supervised RSCD methods in real-world scenarios.

B. Semi-Supervised RSCD

SSL aims to alleviate the issue of high annotation costs by synergistically utilizing a small amount of labeled data and massive amounts of unlabeled data. In the RSCD domain, existing SSL methods can be broadly categorized into three types: generative-based, pseudo-label-based, and consistency regularization-based methods.

To reduce dependency on labeled training datasets, SSL has been introduced into the field of remote sensing image change detection. Generative-based methods typically model unlabeled data via generative models and identify change regions using reconstruction errors, latent space differences, or adversarial training. Gong et al. [25] proposed a generative discriminative classification network, mitigating performance degradation caused by insufficient annotations by jointly training a generator and a classifier. S4GAN [12] introduced a self-training strategy based on discriminator scores, utilizing high-confidence predictions from the generator as pseudo-labels to significantly improve change detection performance under semi-supervised conditions. Pseudo-label-based methods expand the training set size by using model predictions on unlabeled samples as supervisory signals. Ran et al. [14] proposed AdaSemiCD, which introduces quantifiable pseudo-label quality metrics and enhances pseudo-label reliability through class reweighting and ambiguous region enhancement. FixMatch [26] employs thresholding to filter high-confidence predictions and combines weak augmentation predictions with strong augmentation consistency losses for training, rendering pseudo-label learning more stable and efficient. Consistency regularization-based methods enforce model output consistency by applying different perturbations to unlabeled samples, thereby improving model robustness. Shu et al. [27] adopted change-aware CutMix data augmentation to construct the CutMix-CD framework, enabling the model to maintain sensitivity to change regions under various data perturbations. Hafner et al. [28] proposed the dual-task SSCD framework, enhancing joint learning capabilities by constructing consistency constraints between segmentation and change detection. Zhang et al. [29] employed dual constraints of category consistency and feature consistency, enabling the model to learn feature representations closer to the true change distribution within the semantic space of unlabeled samples.

Although existing SSL-CD methods have made certain progress, they still suffer from the following limitations. Most methods directly utilize standard backbone networks, failing to fully mine the structural supervision embedded in sparse labels, resulting in limited representation capabilities for complex boundaries and multiscale changes. Meanwhile, many methods use pseudo-labels indiscriminately; high-noise pseudo-labels lead to error accumulation and confirmation bias, severely affecting model convergence quality and final performance. To address this issue, this article proposes a semi-supervised change detection framework for remote sensing imagery that combines adaptive confidence filtering with attention-guided feature aggregation, aiming to improve pseudo-label quality, enhance feature expression capabilities, and thereby elevate the overall performance of semi-supervised RSCD.

III. Methodology

A. Overview of the AG-SemiCD Framework

AG-SemiCD is a semi-supervised change detection framework designed based on the mean teacher paradigm [30]. The framework aims to alleviate the dependence on labeled data by exploiting the latent feature distributions embedded in large-scale unlabeled data. The overall architecture is illustrated in Fig. 1. Both the teacher network and the student network adopt the proposed AGNet architecture. The teacher network is responsible for generating pseudo-labels, while the student network is trained jointly using labeled data and pseudo-labels, with its parameters being continuously updated. The output of the teacher network guides the learning process of the student network, and parameter updates from the student network are subsequently fed back to the teacher network, forming a self-consistent learning loop. The training process consists of a supervised stage and a semi-supervised stage.

During the supervised learning stage, paired remote sensing images and their corresponding ground truth labels are

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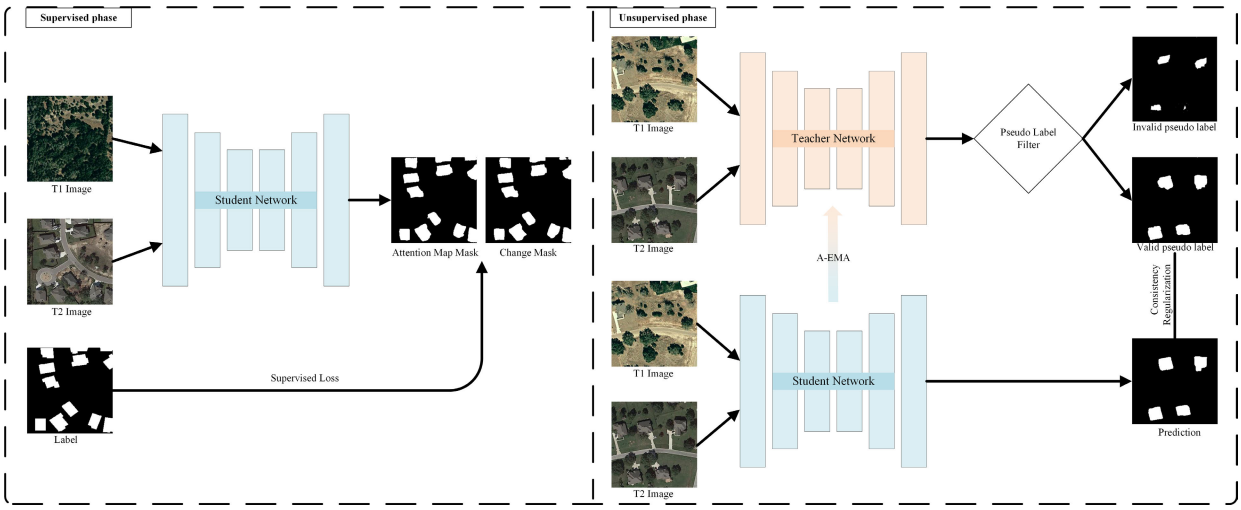


Fig. 1. Overall framework of AG-SemiCD consists of a supervised stage and an unsupervised stage.

provided as inputs. The student network performs feature extraction and prediction on the input images, and explicitly learns accurate change representations by minimizing the supervised loss between the predictions and the ground-truth labels, thereby establishing the fundamental discriminative capability of the model.

During the SSL stage, large-scale unlabeled images are used as inputs. The unlabeled data are first fed into the teacher network to generate preliminary prediction maps. To suppress noise accumulation, a confidence-aware pseudo-label filtering module is designed, which strictly removes low-quality predictions by leveraging uncertainty estimation and spatial constraints, retaining only high-confidence regions as valid pseudo-labels. The same batch of unlabeled data is also fed into the student network. An unsupervised loss is computed between the student predictions and the valid pseudo-labels generated by the teacher network, enforcing prediction consistency under different perturbations or augmentations and enabling the model to capture the intrinsic structure of the data. Meanwhile, an A-EMA mechanism is employed to dynamically update the parameters of the teacher network according to the state of the student network, ensuring the stability and reliability of pseudo-labels throughout the entire training process.

1) Pseudo-Label Filter: In semi-supervised change detection tasks, the quality of pseudo-labels generated by the teacher network is paramount. Conventional methods often utilize all pseudo-labels indiscriminately for training, which inevitably introduces a substantial amount of noise and erroneous supervisory signals, resulting in error accumulation during the training process. To address this issue, a confidence-aware pseudo-label filter is proposed. By leveraging uncertainty estimation and spatial distribution constraints, this module rigorously selects high-quality pseudo-labels from both pixel-level and image-level dimensions.

In RSCD tasks, the inherent sparsity and spatial heterogeneity of changed regions make pseudo-label noise highly prone to interfering with model convergence. To accurately assess prediction reliability, information entropy is introduced as a core metric to quantify the uncertainty of the teacher network's predictions, effectively characterizing the overall dispersion of the predicted probability distribution. On this basis, a dynamic thresholding strategy is proposed: a strict criterion is adopted during the early stage of training to ensure optimization stability, and the filtering threshold is gradually relaxed as model confidence improves. This strategy effectively suppresses noise accumulation while significantly enhancing the utilization of unlabeled data in the later training stages, thereby achieving a balance between stability and data efficiency. Let $\tilde{Y}^i \in \mathbb{R}^{H\times W}$ denote the prediction map generated by the teacher network for the i unlabeled sample. For a pixel at spatial location (j,k), the predicted probability is defined as $p^i_{jk} = \sigma(\tilde{y}^i_{jk})$, where $\sigma(x) = 1/(1+e^{-x})$ denotes the sigmoid function. The entropy $H(p^i_{jk})$ of this pixel is as follows:

$$H\left(p^i_{jk}\right) = -p^i_{jk}\log\left(p^i_{jk}+\epsilon\right) - \left(1-p^i_{jk}\right)\log\left(1-p^i_{jk}+\epsilon\right) \quad (1)$$

where $\epsilon = 10^{-8}$ is a numerical stability constant introduced to prevent logarithmic overflow.

To evaluate sample quality from a global perspective, the image-level confidence score $\mathrm{Conf}(\tilde{Y}^i)$ is defined as the complement of the average entropy over the spatial dimensions. A higher value indicates stronger overall prediction certainty of the teacher network for the current sample

$$\mathrm{Conf}\left(\tilde{Y}^i\right) = 1 - \frac{1}{H\times W}\sum_{j=1}^{H}\sum_{k=1}^{W} H\left(p^i_{jk}\right). \quad (2)$$

Considering the progressive learning behavior of the model from weak to strong during training, a fixed confidence threshold is insufficient to adapt to different training stages. Therefore, a dynamic confidence threshold τ_{conf} is introduced. At early training stages, a high threshold is employed to ensure that only the most reliable pseudo-labels participate in gradient updates, preventing noise interference during the cold-start phase. As training proceeds, the threshold is linearly decreased

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to exploit a broader range of unlabeled samples. The threshold is defined as

$$\tau_{\text{conf}} = \max\left(0.95 - 0.15 \times \frac{e}{E_{\text{total}}},\ 0.8\right) \tag{3}$$

where E_{total} denotes the total number of training epochs and e represents the current epoch.

Given that change regions in RSCD are typically sparse, class imbalance is a common issue. To further filter abnormal samples, a change ratio constraint is introduced. The proportion of changed pixels in sample i, denoted as R^i_{change}, is computed as

$$R^i_{\text{change}} = \frac{1}{H \times W} \sum_{j=1}^{H} \sum_{k=1}^{W} \mathbb{I}\left[p^i_{jk} > 0.5\right]. \tag{4}$$

Extreme change ratios (e.g., nearly all-black or all-white masks) are often associated with imaging anomalies, invalid data, or severe domain shifts. Therefore, a pseudo-label is included in the valid set M_{valid} only when it simultaneously satisfies both the confidence constraint and the spatial distribution constraint

$$M_{\text{valid}} = \left\{i \mid \text{Conf}\left(\tilde{Y}^i\right) > \tau_{\text{conf}} \ \wedge\ R_{\min} < R^i_{\text{change}} < R_{\max}\right\} \tag{5}$$

where $R_{\min} = 0.01$ and $R_{\max} = 0.5$ are empirically set to ensure the semantic plausibility of the pseudo-labels.

2) Adaptive Exponential Moving Average: In the standard teacher–student framework, the parameters of the teacher network $\boldsymbol{\theta}_t$ are typically updated by applying an EMA over the parameters of the student network $\boldsymbol{\theta}_s$. The conventional EMA [30] update rule is defined as

$$\boldsymbol{\theta}'_t = \alpha \boldsymbol{\theta}'_{t-1} + (1 - \alpha)\, \boldsymbol{\theta}_t \tag{6}$$

where $\boldsymbol{\theta}'_t$ and $\boldsymbol{\theta}'_{t-1}$ denote the teacher network parameters at iteration t and $t-1$, respectively, $\boldsymbol{\theta}_t$ represents the student network parameters at iteration t, and α is a fixed momentum coefficient, typically set to 0.99.

However, a fixed momentum coefficient α is insufficient to accommodate the nonstationary dynamics of the training process. At early training stages, when the student network has not yet converged, a fixed low α may cause the teacher network to update too rapidly, making it prone to absorbing noisy and unstable features. Conversely, at later stages when the student network becomes more mature, a fixed high α leads to overly slow updates of the teacher network, preventing it from effectively incorporating fine-grained feature improvements learned by the student.

To address this issue, we propose an adaptive EMA strategy (A-EMA), which dynamically adjusts the momentum coefficient α_t according to the real-time performance of the student network. Specifically, the prediction confidence of the student network, denoted as c_t, is used as a proxy indicator of its training state

$$c_t = 1 - \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} H\left(p^t_{i,j,\text{student}}\right) \tag{7}$$

where $H(\cdot)$ denotes the information entropy and $p^t_{i,j,\text{student}}$ represents the predicted probability of the student network at pixel location (i, j) at iteration t.

Based on c_t, the momentum coefficient α_t is adaptively interpolated between a minimum value $\alpha_{\min}$ and a maximum value $\alpha_{\max}$

$$\alpha_t = \alpha_{\min} + (\alpha_{\max} - \alpha_{\min}) \times c_t. \tag{8}$$

Following the mean teacher paradigm [30], $\alpha_{\min} = 0.99$ and $\alpha_{\max} = 0.9999$ are adopted. When the student network approaches convergence ($c_t \to 1$), α_t gradually approaches $\alpha_{\max}$. In this case, the teacher network updates at a very slow rate, which helps smooth noise, stabilize pseudo-label generation, and prevent model collapse. Conversely, when the student network is in an uncertain training stage ($c_t \to 0$), α_t approaches $\alpha_{\min}$, allowing the teacher network to update more rapidly so as to quickly adapt to and absorb newly learned features from the student network.

Overall, the proposed A-EMA strategy effectively balances the parameter evolution speed throughout the entire semi-supervised training process, ensuring that the teacher network consistently provides stable and high-quality supervisory signals.

B. AGNet

To address the insufficient utilization of multiscale features and the interference of complex background noise in remote sensing image change detection, this article proposes a Siamese attention-guided change detection network named AGNet. As illustrated in Fig. 2, AGNet is designed to fully exploit multiscale structural features and semantic change patterns from bitemporal remote sensing images, enabling high-precision and fine-grained change detection. The network adopts EfficientNet-B2 [31] with shared parameters as the backbone feature extractor, ensuring that bitemporal images are encoded in the same feature space and reducing noise caused by structural discrepancies.

First, the input bitemporal images are passed through the backbone modules of EfficientNet-B2 at various levels to extract semantic information at different hierarchies. Specifically, shallow features from block1, block2, and block3 retain rich spatial texture details and are particularly sensitive to minute structural changes. Deep features from block5 contain more abstract semantic information, capable of characterizing high-level change patterns. To maintain feature scale consistency and stability, block4 is excluded from subsequent fusion processing because its spatial resolution is identical to that of block5.

Subsequently, shallow and deep features are fed into the Dual-Stream attention perception fusion module (APFM), respectively. This module achieves cross-temporal feature alignment and difference enhancement to extract multiscale change semantics. The deep change features are further processed by the AMG to construct a semantic attention map, which highlights potential change regions. The generated high-semantic attention map is upsampled and then input into the attention-guided refinement module (AGRM). This module refines the shallow multiscale features by suppressing regional noise and emphasizing local structural changes.

Finally, the multiscale change features optimized by AGRM are input into a multiscale refinement fusion head to complete

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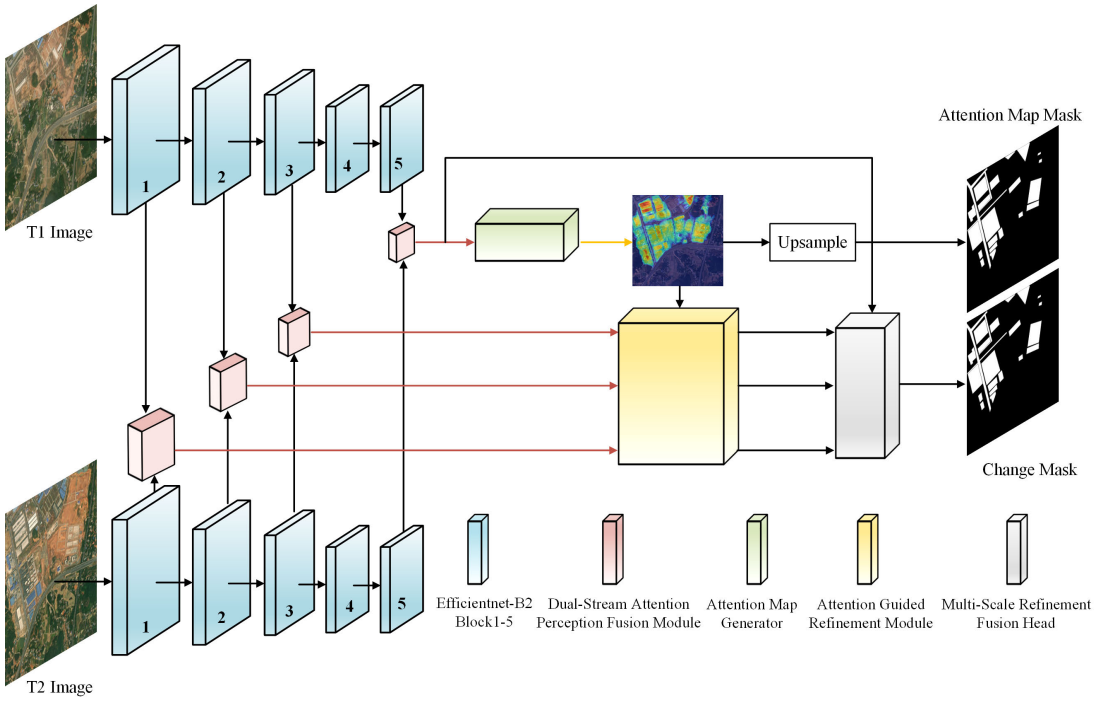


Fig. 2. Structure diagram of AGNet: AGNet takes a pair of images as input and outputs a change mask and an Attention map mask. It consists of five parts: efficientnet-B2, dual-stream attention perception fusion module, AMG, attention-guided refinement module, and multiscale refinement fusion head.

cross-scale feature aggregation and generate the final change mask prediction. Simultaneously, the semantic attention map is upsampled to generate an attention mask, providing additional saliency cues for the model.

Overall, AGNet constructs a framework that integrates multiscale attention mechanisms. Through the synergistic interaction between semantic and spatial features, it achieves a hierarchical representation that progressively evolves from coarse to fine and from deep to shallow. Specifically, APFM focuses on cross-temporal feature modeling, AMG is dedicated to building semantic saliency guidance, and AGRM is responsible for the fine-grained optimization of shallow features under semantic guidance. These three modules complement one another, jointly establishing a rigorous top–down hierarchical semantic architecture.

1) Dual-Stream Attention Perception Fusion Module: The dual-stream APFM is designed to interactively fuse deep semantic features and shallow structural features from bitemporal remote sensing images, modeling change patterns at the same geographical locations across different observation times from both explicit and implicit perspectives. APFM is applied to the output features of block1, block2, block3, and block5 of EfficientNet-B2, constructing a bottom–up multiscale change semantic representation that provides rich hierarchical information for subsequent change interpretation.

As shown in Fig. 3. First, to eliminate channel-wise discrepancies between cross-temporal features, APFM employs a 1×1 convolution to project features into a unified representation space. Subsequently, based on the CBAM [32], a lightweight channel–spatial joint attention mechanism is introduced to refine the features. Channel attention adaptively reweights feature channels by modeling global semantic

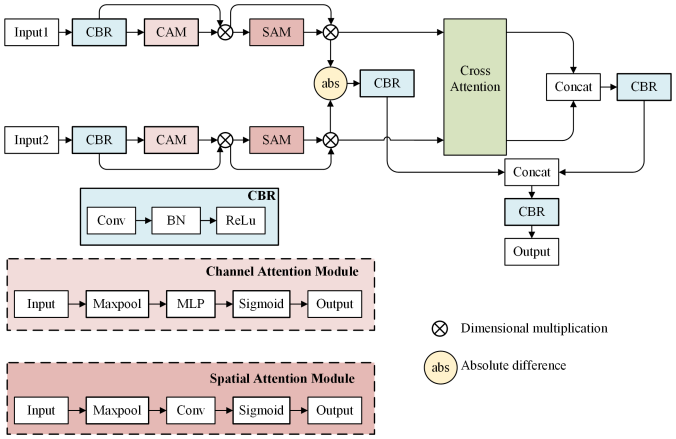


Fig. 3. Structure of the APFM.

dependencies, thereby emphasizing semantic dimensions that are more sensitive to change discrimination. Spatial attention further focuses on potential change regions, enabling the model to suppress background interference and fine-grained noise. After the sequential application of the two attention mechanisms, the refined features are obtained via element-wise multiplication, ensuring that the features possess strong change sensitivity prior to fusion.

To jointly account for explicit change magnitude and cross-temporal contextual correlation, APFM incorporates two complementary interaction streams: a difference stream and a cross-attention stream.

Difference Stream: The enhanced bitemporal features are subtracted element-wise to directly capture pixel-level intensity differences and local structural changes. This oper-

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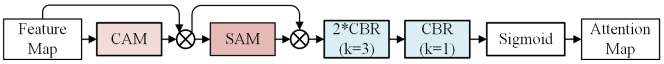


Fig. 4. Structure of the AMG.

ation exhibits strong responsiveness to explicit changes such as building demolition and land-cover morphology variation, serving as a primary source of change evidence.

Cross-Attention Stream: To alleviate feature misalignment caused by slight registration errors, viewpoint variations, or sensor noise, a bidirectional cross-attention mechanism [33] is introduced, allowing features at each temporal point to query contextual information from the other. By dynamically establishing semantic associations between the two temporal representations, the model captures implicit interdependencies and enhances structural consistency, thereby improving the recognition of subtle changes, semantic variations, and occlusion-induced changes.

Finally, the explicit difference features and implicit relational features are fused via convolution to generate a more discriminative and robust multiscale change representation. The resulting fused features not only highlight the saliency of change regions but also preserve cross-scale semantic consistency, laying a solid foundation for subsequent attention-guided refinement and fine-grained change mask prediction.

2) Attention Map Generator: Although deep features exhibit relatively low spatial resolution, they encode the most abundant high-level semantic information, which is crucial for distinguishing pseudo changes from genuine land-cover changes. Motivated by this property, the AMG is designed to exploit deep semantic representations to establish global saliency guidance, thereby assisting shallow features in accurately focusing on potential change regions while preserving fine structural details.

As illustrated in Fig. 4, AMG takes the deep multiscale fused features extracted from Block 5 of EfficientNet-B2 as input. To further purify semantic information and suppress redundant responses, AMG first applies a joint enhancement of channel attention and spatial attention to the deep features. Channel attention reallocates channel-wise importance based on global statistical cues, emphasizing semantic dimensions with strong change discrimination capability. Spatial attention, in turn, highlights key regions that are likely to contain changes, enabling the network to achieve preliminary semantic focusing before feature abstraction.

Subsequently, the enhanced features are processed by a sequence of convolutional modules for further semantic abstraction. This operation effectively compresses noise, enhances structural consistency, and improves the representational capacity of complex change patterns. Finally, a 1×1 convolution is employed to reduce the feature dimension to a single channel, generating the final high-level semantic attention map. This attention map acts as a global semantic guide throughout the network, where the response intensity reflects the potential probability of change at each spatial location. Regions with higher responses typically correspond to high-confidence candidate change areas, thereby providing global prior constraints for the subsequent AGRM.

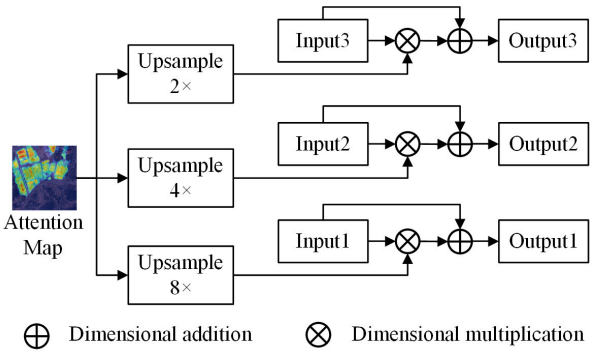


Fig. 5. Structure of the AGRM.

By incorporating AMG, AGNet establishes an effective cross-layer guidance mechanism between deep semantic representations and shallow structural details. This design enables the network to capture large-scale change patterns while maintaining high discriminative power and stability at fine-grained structural levels.

3) Attention-Guided Refinement Module: Although shallow features are advantageous in capturing spatial textures and boundary details, they are often contaminated by substantial background noise and irrelevant nonchange information. Directly leveraging such features for change discrimination may therefore lead to false alarms. To address this issue, we propose an AGRM, which imposes saliency constraints on shallow features using the high-level semantic attention maps generated by AMG, enabling the network to preserve fine details while focusing more effectively on potential change regions.

The architecture of AGRM is illustrated in Fig. 5. First, the single-channel semantic attention map produced from deep features is upsampled via bilinear interpolation to precisely match the spatial resolutions of the shallow features from Block1, Block2, and Block3. This operation ensures pixel-wise alignment between the attention map and shallow features, allowing deep semantic saliency to be propagated into low-level spatial structures at different scales. The upsampled attention map is then used as a weighting mask and multiplied element-wise with the shallow features at the corresponding scale. This process effectively suppresses noise inherent in shallow representations, encouraging the model to allocate greater attention to land-cover structures that are more likely to undergo changes.

To prevent excessive attenuation of fine-grained details, the weighted features are further combined with the original shallow features through a residual enhancement scheme. By integrating semantic-guided weighting with residual connections, AGRM explicitly strengthens change-sensitive regions while preserving intrinsic structural details.

Through this mechanism, AGRM achieves saliency-guided refinement of shallow features, effectively balancing detail preservation and noise suppression, and thereby providing

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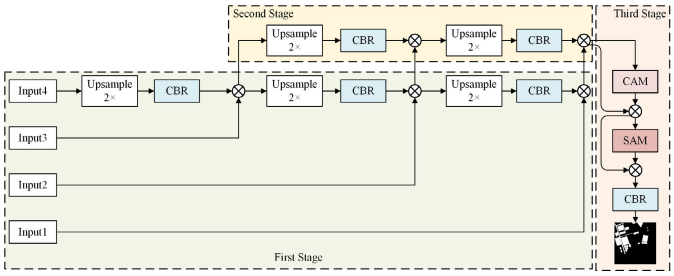


Fig. 6. Structure of the multiscale refinement fusion head.

more discriminative multiscale representations for accurate change detection.

4) Multiscale Refinement Fusion Head: The multiscale refinement fusion head is designed to integrate all multiscale features optimized by the AGRM modules, constructing a structurally consistent and semantically discriminative final change representation and generating the predicted change mask. Since features at different hierarchical levels exhibit complementary properties in terms of spatial details, semantic abstraction, and receptive field sizes, effective multiscale fusion is crucial for improving change detection accuracy.

As illustrated in Fig. 6, the proposed multiscale refinement fusion head consists of three key stages: gradient fusion, cascaded aggregation, and attention-refined output. In the first stage, skip connections and cross-scale fusion strategies are employed to progressively merge features from Block 1, Block 2, Block 3, and Block 5 according to their spatial resolutions, enabling mutual compensation between fine-grained details and high-level semantic information across scales. In the second stage, a progressive upsampling scheme is adopted, where deep features are gradually upsampled and fused with shallower features. This coarse-to-fine aggregation strategy effectively alleviates aliasing artifacts caused by large upsampling factors and facilitates smooth recovery of spatial resolution. In the third stage, the fused features are further refined through channel attention and spatial attention mechanisms, followed by a convolutional prediction to generate the final change mask.

Through hierarchical aggregation and attention-based refinement, the multiscale refined fusion head effectively consolidates complementary information from multiple feature levels, yielding a robust and precise representation for accurate change detection.

C. Loss Functions

1) Supervised Learning Loss: In the supervised learning stage, paired input images (A, B) with the corresponding ground-truth label $Y \in \{0,1\}^{H\times W}$ are provided. After being processed by AGNet, the network outputs a change detection prediction $\hat{Y}_c = f_c(A,B)$ and an auxiliary attention map prediction $\hat{Y}_f = f_f(A,B)$. Calculate the loss for a single pixel (i, j) using the binary cross-entropy loss [34]

$$\mathcal{L}_{\mathrm{BCE}}\left(y_{ij},\hat{y}_{ij}\right) = -\left[y_{ij}\log\left(\sigma\left(\hat{y}_{ij}\right)\right) + \left(1-y_{ij}\right)\log\left(1-\sigma\left(\hat{y}_{ij}\right)\right)\right] \quad (9)$$

where $\sigma(x) = 1/(1+e^{-x})$ denotes the sigmoid function and $\hat{y}_{ij}$ is the logit output.

For the labeled dataset D_l, the overall supervised loss $\mathcal{L}_{\mathrm{sup}}$ is defined as

$$\mathcal{L}_{\mathrm{sup}} = \frac{1}{|D_l|}\sum_{(A,B,Y)\in D_l}\left[\mathcal{L}_{\mathrm{BCE}}\left(\hat{Y}_c, Y\right) + \mathcal{L}_{\mathrm{BCE}}\left(\hat{Y}_f, Y\right)\right]. \quad (10)$$

2) SSL Loss: In the SSL stage, for an unlabeled image pair (A_u, B_u), the teacher network produces the predictions $\tilde{Y}_c = \sigma(f_c^{\mathrm{teacher}}(A_u, B_u))$ and $\tilde{Y}_f = \sigma(f_f^{\mathrm{teacher}}(A_u, B_u))$. For the filtered valid sample set M_{valid}, a pseudo-label supervision loss and a consistency regularization loss are jointly defined.

The pseudo-label supervision loss $\mathcal{L}_{\mathrm{pseudo}}$ is computed only on high-confidence samples

$$\mathcal{L}_{\mathrm{pseudo}} = \frac{1}{|M_{\mathrm{valid}}|}\sum_{i\in M_{\mathrm{valid}}}\left[\mathcal{L}_{\mathrm{BCE}}\left(\hat{Y}_c^{(i)}, \tilde{Y}_c^{(i)}\right) + \mathcal{L}_{\mathrm{BCE}}\left(\hat{Y}_f^{(i)}, \tilde{Y}_f^{(i)}\right)\right]. \quad (11)$$

To further enforce consistency between the student and teacher networks, we introduce a logits consistency loss $\mathcal{L}_{\mathrm{cons_mse}}$ and an edge consistency loss $\mathcal{L}_{\mathrm{cons_edge}}$.

The logits consistency loss is defined using the mean squared error

$$\mathcal{L}_{\mathrm{cons_mse}} = \frac{1}{|M_{\mathrm{valid}}|HW}\sum_{i\in M_{\mathrm{valid}}}\sum_{j,k}\left(\hat{y}_{f,jk}^{(i)} - \tilde{y}_{f,jk}^{(i)}\right)^2. \quad (12)$$

Given the boundary-sensitive nature of change detection, geometric edge features are extracted using the Sobel operator. The edge extraction function $E(\cdot)$ is defined as

$$E(p) = \sqrt{G_x^2 + G_y^2 + \varepsilon} \quad (13)$$

where G_x and G_y denote the standard Sobel kernels and ε is a small constant for numerical stability.

The edge consistency loss is formulated as

$$\mathcal{L}_{\mathrm{cons_edge}} = \frac{1}{|M_{\mathrm{valid}}|HW}\sum_{i\in M_{\mathrm{valid}}}\sum_{j,k}\left|E\left(\sigma\left(\hat{Y}_f^{(i)}\right)\right)_{jk} - E\left(\sigma\left(\tilde{Y}_f^{(i)}\right)\right)_{jk}\right|. \quad (14)$$

The overall consistency regularization loss is defined as

$$\mathcal{L}_{\mathrm{cons}} = w_{\mathrm{mse}}\cdot\mathcal{L}_{\mathrm{cons_mse}} + w_{\mathrm{edge}}\cdot\mathcal{L}_{\mathrm{cons_edge}}. \quad (15)$$

Here, w_{mse} and w_{edge} are weighting coefficients used to regulate overall prediction consistency and enforce boundary structural constraints, respectively. To prevent noise gradients caused by over-emphasizing boundary constraints, we set $w_{\mathrm{mse}} = 0.1$ and $w_{\mathrm{edge}} = 0.05$.

Finally, the total loss for the SSL stage is given by

$$\mathcal{L}_{\mathrm{semi}} = \mathcal{L}_{\mathrm{pseudo}} + \mathcal{L}_{\mathrm{cons}}. \quad (16)$$

IV. Experiments

A. Datasets and Evaluation Metrics

To comprehensively evaluate the effectiveness and generalization capability of the proposed AG-SemiCD framework, we conduct systematic experiments on three representative RSCD

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TABLE I
DETAILED COMPARISON OF THE THREE DATASETS

Dataset	Image pairs	Image size	Train/Val/Test	Changes	Resolution
LEVIR-CD	637	1024×1024	7120/1024/2048	building	0.5
WHU-CD	1	32507×15354	5376/768/1536	building	0.2
MSRS-CD	842	1024×1024	9408/1344/2704	building,planting,road,etc	0.5

benchmark datasets from the perspectives of change semantic diversity and remote sensing scene complexity, including LEVIR-CD [6], WHU-CD [35], and multiscale remote sensing change detection dataset (MSRS-CD) [36]. Detailed descriptions of these datasets are provided as follows.

LEVIR-CD was released by Beihang University in 2020 and is a public benchmark dataset specifically designed for building change detection. It consists of 637 pairs of bitemporal remote sensing images with a spatial resolution of 0.5 m, where each image pair has a spatial size of 1024 × 1024 pixels. The images were collected between 2002 and 2018, covering multiple urban areas in Texas, USA. The annotations focus on changes in man-made structures, including high-rise apartments, detached houses, and various auxiliary buildings. Owing to its clear semantic annotations, long temporal span, and diverse architectural styles, LEVIR-CD is well-suited for evaluating model robustness against seasonal variations and illumination differences.

WHU-CD, constructed and released by Wuhan University in 2019, is one of the highest-resolution and largest-coverage change detection benchmark datasets to date. It is based on aerial imagery acquired over Christchurch, New Zealand, in 2012 and 2016, and provides a pair of panchromatic images with a spatial resolution of 0.2 m. The original image size reaches 32 207 × 15 354 pixels, covering an area of approximately 20.5 km^2. The dataset contains fine-grained annotations of change states for 12 796 buildings, making it particularly suitable for high-precision, large-scale urban change detection studies and for validating algorithm robustness and generalization performance in complex urban environments.

MSRS-CD, released by Changsha University of Science and Technology in 2024, is designed to address the challenges of detecting multitype and multiscale land-cover changes in real-world scenarios. The dataset includes 841 pairs of bitemporal remote sensing images with a spatial resolution of 0.5 m, and each image pair has a size of 1024×1024 pixels. The data were collected from several rapidly urbanizing regions in southern China between 2019 and 2023. In addition to building changes, MSRS-CD covers a wide range of change types such as road construction, vegetation dynamics, and suburban expansion, with pixel-level change annotations provided. Owing to its high scene heterogeneity and rich change semantics, MSRS-CD offers a rigorous benchmark for comprehensive performance evaluation under complex real-world conditions.

For all datasets, the original images are cropped into patches of 256 × 256 pixels and split into training, validation, and test sets. The detailed dataset partition statistics are summarized in Table I.

This article adopts five common evaluation metrics for change detection algorithms to assess model performance: Precision (P), Recall (R), IoU, and the $F1$ score. Their specific definitions are as follows:

$$Precision = \frac{TP}{(TP + FP)} \tag{17}$$

$$Recall = \frac{TP}{FN + TP} \tag{18}$$

$$IoU = \frac{TP}{TP + FN + FP} \tag{19}$$

$$F1 = \frac{2}{Recall^{-1} + Precision^{-1}} \tag{20}$$

where TP denotes true positives (the number of samples correctly predicted as change), TN true negatives (correctly predicted as no change), FP false positives (predicted as change but actually no change), and FN false negatives (predicted as no change but actually change).

B. Implementation Details

The AG-SemiCD framework is implemented based on the PyTorch deep learning platform, with both the student and teacher models instantiated using AGNet. During training, the student network is optimized using the AdamW optimizer with an initial learning rate of 0.0005, a weight decay of 0.0025, and a batch size of 32. All experiments are conducted on a single NVIDIA Tesla L40 GPU with 48 GB of memory to ensure consistent computational resources. The total training process spans 200 epochs, where the first 15 epochs employ purely supervised training using only labeled data to stabilize the model. After this warm-up stage, unlabeled data are introduced to initiate the semi-supervised training phase. To enhance model generalization and mitigate overfitting, data augmentation is applied, including geometric transformations such as flipping, rotation, cropping, and brightness adjustment. All comparison methods employ officially released implementation versions or configuration schemes. Dataset partitioning and evaluation metrics remain consistent across all methods. When strict uniformity proves unfeasible due to architectural differences, each method is evaluated under its recommended settings to ensure fair and representative performance comparisons.

C. Comparative Experiments With SOTA Algorithms

To comprehensively evaluate the performance of the proposed algorithm, we selected eight SOTA algorithms for comparison on the LEVIR-CD, WHU-CD, and MSRS-CD datasets. These algorithms include s4GAN [12], AdvNet [37], SemiCDNet [38], SemiCD [39], RCL [15], Unimatch-P [40], Unimatch-D [40], and C2F-SemiCD [16]. Additionally, two baseline experimental setups were conducted: Sup-only and

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TABLE II
COMPARISON WITH OTHER SOTA MODELS ON THE LEVIR-CD DATASET, WHERE THE BOLDED VALUES ARE THE BEST

Method	Labelled Ration															
	5%				10%				20%				40%			
	F1	P	R	IoU	F1	P	R	IoU	F1	P	R	IoU	F1	P	R	IoU
Sup-only	87.61	91.41	84.11	77.95	89.07	91.25	86.99	80.29	90.34	92.47	88.30	82.37	90.99	92.97	89.10	83.47
s4GAN	77.15	82.72	72.28	62.80	82.19	87.41	77.55	69.76	84.37	89.18	80.06	72.97	86.34	89.43	83.46	75.96
AdvNet	79.94	86.72	74.14	66.58	84.11	89.17	79.60	72.58	85.49	89.72	81.64	74.66	86.17	88.93	83.58	75.70
SemiCDNet	80.56	86.93	75.06	67.45	83.70	89.73	78.44	71.97	85.34	90.22	80.95	74.43	86.09	89.50	82.92	75.57
SemiCD	84.86	89.08	81.03	73.71	85.63	89.25	82.29	74.87	86.36	89.53	83.40	75.99	87.12	88.57	85.73	77.19
RCL	84.70	87.65	81.94	73.45	86.69	89.34	84.20	76.51	87.03	86.97	87.08	77.03	87.64	87.17	88.12	78.00
Unimatch-P	81.62	81.74	81.50	68.95	82.79	84.23	81.39	70.63	83.66	86.09	81.35	71.90	83.67	83.93	83.43	71.93
Unimatch-D	88.62	**93.73**	84.05	79.57	89.29	**93.75**	85.23	80.65	89.73	92.77	86.89	81.38	89.59	93.94	85.63	81.15
C2F-SemiCD	88.89	90.66	87.19	80.00	90.68	92.14	89.27	82.96	91.11	**92.88**	89.40	83.67	91.58	**93.98**	89.30	84.47
AG-SemiCD	**91.14**	92.49	**89.83**	**83.72**	**91.41**	93.20	**89.69**	**84.18**	**91.48**	92.56	**90.43**	**84.30**	**91.61**	92.65	**90.60**	**84.53**
Fully-sup	F1=91.78,P=93.48,R=90.15,IoU=84.82,OA=99.18															

Fully-sup. Sup-only trains the model using only the available labeled data while completely ignoring unlabeled data, representing a baseline performance under data-scarce conditions. Fully-sup trains using all available labeled data in the training set, representing the upper bound performance under ideal data conditions. Brief descriptions of the eight comparison methods are provided below.

1) *s4GAN:* A classic semi-supervised segmentation method. It employs a dual-branch network structure that integrates multitask learning with a self-training mechanism, aiming to mitigate feature artifacts caused by label scarcity through adversarial loss.
2) *AdvNet:* A semi-supervised method based on fully convolutional adversarial networks. It enhances performance by discriminating between predicted probability maps and the ground truth distribution, leveraging trustworthy regions in unlabeled images to provide additional supervision.
3) *SemiCDNet:* A GAN-based semi-supervised change detection method. By introducing two discriminators, it performs feature distribution alignment on both segmentation maps and entropy maps for labeled and unlabeled data, utilizing unlabeled data to boost the model's generalization capability.
4) *SemiCD:* A consistency-regularization-based semi-supervised change detection method. Random perturbations are applied to deep feature difference maps, enforcing consistent output probability maps for unlabeled bitemporal image pairs.
5) *RCL:* A semi-supervised RSCD method based on reliable contrastive learning. Contrastive loss is designed specifically for change regions to strengthen feature extraction, and uncertainty-based filtering selects reliable pseudo-labels from unlabeled data, effectively improving model performance.
6) *Unimatch-P:* Introduces the weak-to-strong consistency framework from FixMatch. Auxiliary feature perturbations expand the perturbation space, and a dual-strong-view structure shares supervision from a single weak view, significantly enhancing performance.
7) *Unimatch-D:* An improved Unimatch variant based on DeepLabv3 + backbone. Dual strong perturbations and channel dropout are employed to enhance consistency regularization, improving pseudo-label stability and boundary detection of change regions.
8) *C2F-SemiCD:* A coarse-to-fine semi-supervised framework. C2FNet progressively refines multiscale features via attention mechanisms and combines a mean teacher architecture to stabilize student model updates using EMA.

Table II presents the quantitative comparison results on the LEVIR-CD dataset under different labeled data ratios (5%, 10%, 20%, and 40%). The analysis indicates that AG-SemiCD outperforms existing semi-supervised change detection methods in key evaluation metrics, namely $F1$ and IoU. Notably, under the extremely challenging 5% labeling ratio, AG-SemiCD demonstrates remarkable data mining capability, achieving an $F1$ score of 91.14% and IoU of 83.72%, which represent improvements of 3.53% and 5.77% over the supervised baseline Sup-only, respectively. Furthermore, AG-SemiCD surpasses the second-best model, C2F-SemiCD, by a significant 2.25% $F1$ margin. This improvement is mainly attributed to a substantial increase in recall, indicating that AG-SemiCD effectively captures weak changes and fine-grained structural details. Moreover, AG-SemiCD exhibits high data utilization efficiency. With only 10% of labeled data, it already achieves an $F1$ score of 91.41%, approaching the performance upper bound of the fully supervised model (91.78%). Compared to GAN-based methods and consistency-regularization-based approaches, AG-SemiCD consistently outperforms across all metrics. This advantage primarily stems from its adaptive confidence-based pseudo-label selection, which suppresses the propagation of noisy pseudo-labels, and its attention-guided feature aggregation, which mitigates cross-temporal feature misalignment, thereby achieving an optimal balance between reduced labeling dependency and high-precision detection. To provide a more intuitive assessment of model performance, Fig. 7 presents qualitative visual comparisons under a 10% labeling ratio. Fig. 7(a) and (b) demonstrates that AG-SemiCD is more effective in reducing false detections and suppressing background noise. Fig. 7(c)–(e) further illustrates its superior capability in boundary delineation, small-object recognition, and fine structural change capture compared to other mod-

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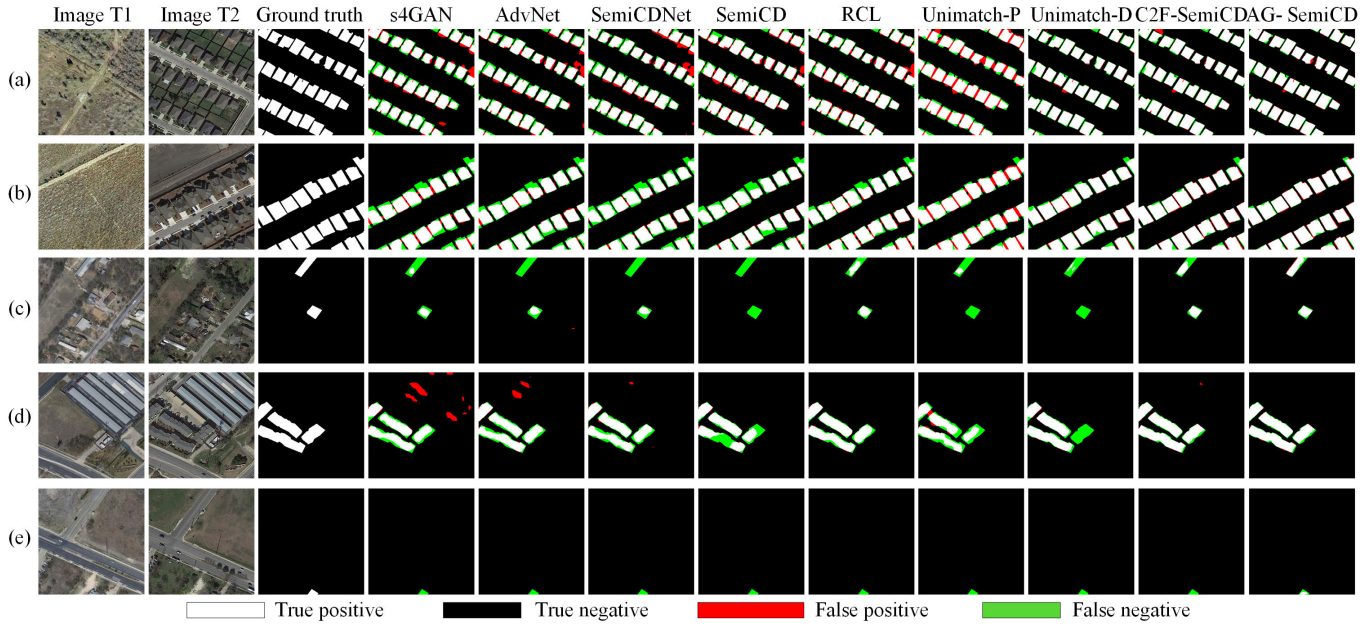


Fig. 7. Visual results of 11 algorithms tested on the LEVIR-CD dataset. Among them, (a)–(e) detection results on LEVIR-CD.

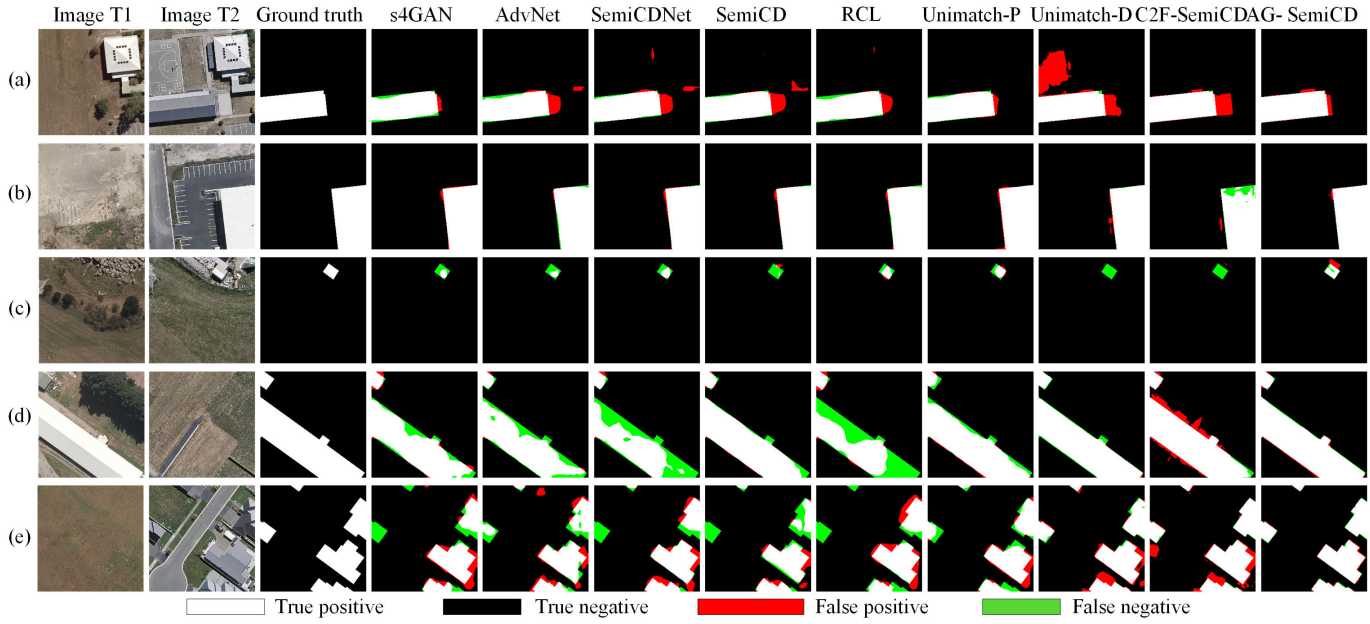


Fig. 8. Visual results of 11 algorithms tested on the WHU-CD dataset. Among them, (a)–(e) detection results on WHU-CD.

els, producing more precise and coherent change region predictions.

Table III presents the experimental results on the WHU-CD dataset under different labeled data ratios (5%, 10%, 20%, and 40%). The results demonstrate that AG-SemiCD consistently achieves the best performance across all label ratio settings, exhibiting highly stable and strong generalization capabilities. Notably, under the extremely low 5% labeling ratio, AG-SemiCD performs exceptionally well, achieving an $F1$ score of 90.49%, which represents an improvement of 4.87% over the supervised baseline Sup-only and significantly outperforms mainstream semi-supervised methods such as s4GAN and SemiCDNet. This indicates that the proposed framework can effectively suppress the performance degradation caused by noisy pseudo-labels under low-resource conditions. Furthermore, AG-SemiCD shows high data utilization efficiency and fast convergence. With only 10% labeled data, it achieves an $F1$ score of 93.55%, reaching 99% of the fully supervised upper bound (94.53%). This result demonstrates that the attention-guided feature aggregation and adaptive pseudo-label selection mechanisms can fully exploit structured change information in high-resolution unlabeled remote sensing images, thereby improving learning under sparse label scenarios. Compared to other SOTA methods, AG-SemiCD

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TABLE III

COMPARISON WITH OTHER SOTA MODELS ON THE WHU-CD DATASET, WHERE THE BOLDED VALUES ARE THE BEST

Method	Labelled Ration															
	5%				10%				20%				40%			
	F1	P	R	IoU	F1	P	R	IoU	F1	P	R	IoU	F1	P	R	IoU
Sup-only	85.62	91.58	80.39	74.86	89.79	91.63	88.03	81.47	92.47	93.77	91.20	85.99	93.82	95.20	92.47	88.35
s4GAN	70.27	82.44	61.23	54.16	73.36	70.92	75.97	57.92	78.96	74.07	84.53	65.23	84.51	87.34	81.86	73.18
AdvNet	69.24	76.19	63.45	52.95	76.93	79.60	74.43	62.50	82.29	79.83	84.91	69.91	86.81	86.66	86.97	76.70
SemiCDNet	66.61	74.08	60.51	49.94	76.80	78.12	75.51	62.33	79.94	73.66	87.40	66.59	87.00	88.13	85.90	76.99
SemiCD	76.22	77.18	75.29	61.58	82.24	85.28	79.40	69.83	86.51	86.86	86.16	76.23	87.86	85.30	90.59	78.36
RCL	71.02	84.83	61.07	55.06	77.04	76.97	77.10	62.65	84.42	84.09	84.74	73.04	86.05	85.95	86.14	75.51
Unimatch-P	74.19	91.42	62.43	58.97	83.68	89.65	78.46	71.94	84.79	89.97	80.18	73.60	86.48	90.68	82.65	76.18
Unimatch-D	84.23	91.18	78.27	72.76	86.01	88.45	83.71	75.46	84.19	85.56	82.87	72.70	85.81	85.87	85.76	75.15
C2F-SemiCD	85.64	87.94	83.47	74.89	88.06	90.23	85.99	78.67	89.64	91.10	88.23	81.23	92.49	92.11	**92.88**	86.03
AG-SemiCD	**90.49**	**91.77**	**89.23**	**82.62**	**93.55**	**93.94**	**93.16**	**87.88**	**93.83**	**95.01**	**92.67**	**88.37**	**94.29**	**96.07**	**92.88**	**89.20**
Fully-sup	F1=94.53,P=96.23,R=92.90,IoU=89.63,OA=99.57															

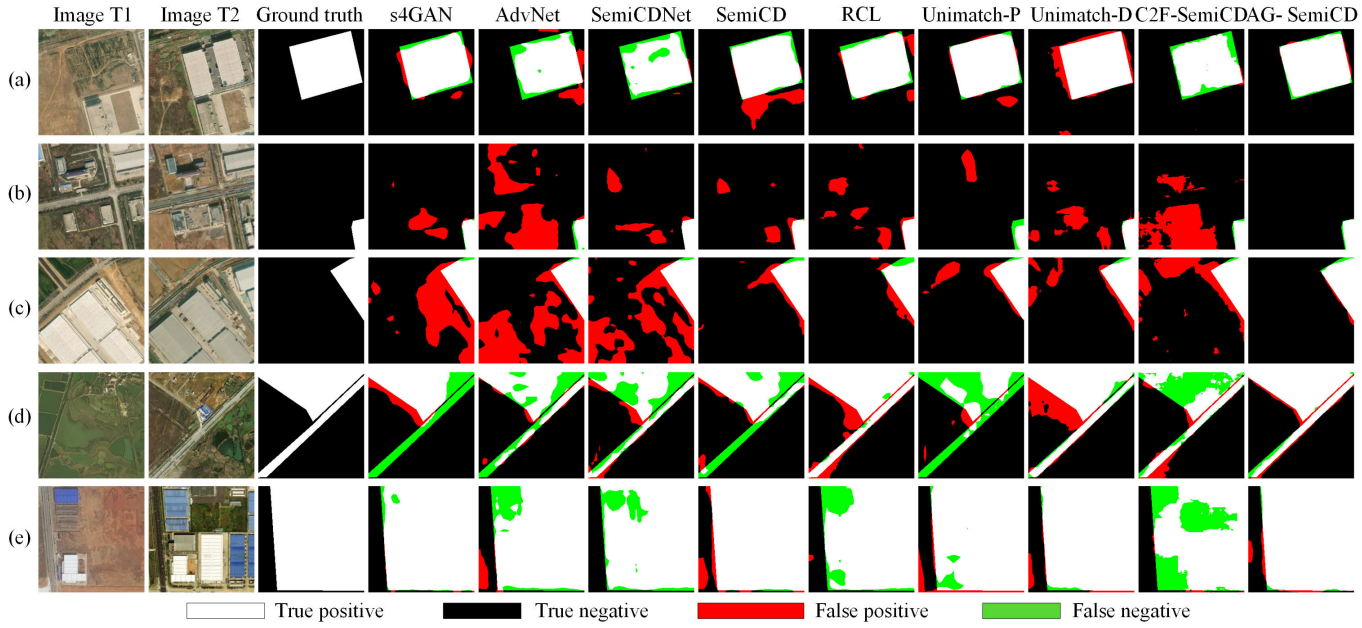


Fig. 9. Visual results of 11 algorithms tested on the MSRS-CD dataset. Among them, (a)–(e) detection results on MSRS-CD.

TABLE IV

COMPARISON WITH OTHER SOTA MODELS ON THE MSRS-CD DATASET, WHERE THE BOLDED VALUES ARE THE BEST

Method	Labelled Ration															
	5%				10%				20%				40%			
	F1	P	R	IoU	F1	P	R	IoU	F1	P	R	IoU	F1	P	R	IoU
Sup-only	72.06	72.50	71.62	56.32	74.21	73.17	**75.29**	59.00	74.49	73.45	**75.57**	59.36	74.75	76.92	72.69	59.68
s4GAN	67.39	70.22	64.78	50.82	68.92	77.27	62.20	52.58	70.40	70.66	70.13	54.32	73.07	74.87	71.35	57.57
AdvNet	64.72	65.50	63.96	47.84	67.30	71.10	63.88	50.71	68.15	76.45	61.48	51.69	70.92	75.11	67.17	54.94
SemiCDNet	66.20	75.30	59.06	49.47	66.40	72.68	61.11	49.70	69.77	70.67	68.89	53.58	71.47	72.61	70.36	55.60
SemiCD	63.61	58.10	70.29	46.64	67.95	71.91	64.41	51.46	70.19	71.65	68.79	54.07	72.79	71.53	74.09	57.22
RCL	71.26	72.33	70.41	55.35	72.13	73.94	70.41	56.41	73.49	73.49	73.48	58.09	74.38	70.55	**78.64**	59.21
Unimatch-P	70.16	70.65	69.69	54.04	70.26	74.99	66.09	54.15	71.44	76.15	67.29	55.57	72.04	**80.67**	65.09	56.30
Unimatch-D	71.30	72.25	70.37	55.40	72.20	71.68	72.72	56.49	72.84	74.87	70.92	57.28	73.40	78.26	69.11	57.97
C2F-SemiCD	71.66	72.52	**70.81**	55.83	72.66	72.52	72.81	57.02	73.74	72.79	74.71	58.40	74.20	73.43	74.99	58.99
AG-SemiCD	**73.36**	**76.53**	70.45	**57.93**	**74.72**	**77.83**	71.85	**59.64**	**75.14**	**77.60**	72.84	**60.19**	**75.47**	77.64	73.42	**60.61**
Fully-sup	F1=75.51,P=78.70,R=72.57,IoU=60.66,OA=93.05															

exhibits superior detection accuracy and structural consistency, particularly in preserving building edges and identifying small change regions with higher robustness. The qualitative visualizations in Fig. 8 further validate these findings. AG-SemiCD significantly reduces both false positives and false negatives compared to competing methods, highlighting its excellent change representation capability in complex urban scenarios.

Table IV presents the experimental results of AG-SemiCD on the MSRS-CD dataset under different labeled data ratios (5%, 10%, 20%, and 40%). Compared to LEVIR-CD and

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TABLE V

ANALYSIS RESULTS OF DIFFERENT MODEL COMPLEXITIES USING 10% LABELED DATA ON THE WHU-CD DATASET

Methods	F1	IoU	FLOPs(G)	Param(M)	Model size
s4GAN	73.36	57.92	73.23	46.85	189.63
AdvNet	76.93	62.50	73.23	46.85	189.62
SemiCDNET	76.80	62.33	73.23	46.85	200.11
SemiCD	82.24	69.83	73.23	46.85	193.74
RCL	77.04	62.65	73.23	46.85	179.08
Unimatch-P	83.68	71.94	24.80	46.71	356.71
Unimatch-D	86.01	75.46	29.10	40.47	309.14
C2F-SemiCD	88.06	78.67	62.10	16.17	62.36
AG-SemiCD	93.55	87.88	12.38	31.34	76.08

TABLE VI

ABLATION EXPERIMENT RESULTS FOR DIFFERENT MODULES OF AGNET USING 10% LABELED DATA ON THE LEVIR-CD AND WHU-CD DATASETS

APFM	AMG	AGRM	LEVIR-CD		WHU-CD	
			F1	IoU	F1	IoU
✗	✓	✓	90.50	82.64	90.34	82.46
✓	✗	✓	88.76	81.62	89.79	81.47
✓	✓	✗	90.33	82.37	91.79	84.18
✓	✓	✓	91.41	84.18	93.55	87.88

WHU-CD, MSRS-CD contains more complex scenes, diverse land-cover categories, and more pronounced scale variations, making it a more challenging benchmark for change detection. Despite these difficulties, AG-SemiCD consistently achieves the best performance across all labeling ratios, demonstrating excellent robustness and cross-scene generalization capability. At the extremely low 5% labeling ratio, AG-SemiCD performs most notably, achieving an $F1$ score of 73.36%, improving by 1.30% over the supervised baseline Sup-only, and outperforming other semi-supervised methods such as Unimatch-D (71.30%) and C2F-SemiCD (71.66%). This result strongly indicates that the adaptive confidence-based pseudo-label selection and attention-guided feature fusion mechanisms of AG-SemiCD can effectively suppress the negative impact of noisy pseudo-labels in complex high-noise scenarios, maintaining stable feature learning. As the labeled ratio increases, AG-SemiCD's advantage further expands. Under the 40% labeling ratio, its $F1$ score reaches 75.47%, with a minimal gap of only 0.04% compared to the fully supervised baseline (75.51%), almost achieving parity. This result strongly demonstrates that AG-SemiCD not only possesses excellent noise suppression capability but can also leverage deep semantic guidance to effectively extract latent features in complex scenarios, achieving high detection accuracy while significantly reducing reliance on manual annotation. The qualitative visualization in Fig. 9 further confirms the superiority of AG-SemiCD. Compared to other methods, AG-SemiCD exhibits lower false-positive and false-negative rates in complex and dynamic scenes, particularly excelling in capturing fine-grained structures such as object boundaries, small-scale targets, and weak change regions.

D. Model Complexity Analysis

Table V compares AG-SemiCD with several representative semi-supervised change detection methods on the WHU-CD dataset under a 10% labeled data setting, evaluating both performance and model complexity. The comparison metrics include $F1$, IoU, floating-point operations (FLOPs), number of parameters, and model size. The experimental results demonstrate that AG-SemiCD achieves extremely high detection accuracy while maintaining a significantly lightweight design, achieving an optimal balance between performance and complexity.

In terms of computational efficiency, AG-SemiCD requires only 12.38 G FLOPs, the lowest among all compared methods, corresponding to 16.9% of s4GAN (73.23 G) and 19.9% of C2F-SemiCD (62.10 G). This computational advantage implies high inference efficiency, making the model suitable for real-time remote sensing processing, rapid disaster assessment, and other time-critical scenarios. Regarding spatial complexity, although C2F-SemiCD slightly outperforms in terms of parameter count (16.17 M), AG-SemiCD achieves a significant improvement in $F1$ score from 88.06% to 93.55% with a moderate increase in parameters to 31.34 M, demonstrating excellent parameter utilization efficiency. Furthermore, compared with the parameter-heavy Unimatch series (>40 M parameters, >300 MB model size), AG-SemiCD achieves superior detection performance with a compact model size of 76.08 MB, making it more suitable for deployment on resource-constrained platforms.

In summary, AG-SemiCD demonstrates the lowest computational cost while delivering SOTA-level detection accuracy, highlighting its practical value as an efficient and robust benchmark model for semi-supervised change detection.

V. ABLATION STUDIES

A. Ablation of AGNet Modules

To comprehensively evaluate the effectiveness of each key component in AGNet, ablation experiments were conducted on the LEVIR-CD and WHU-CD datasets using 10% labeled data, and the results are reported in Table VI. It should be noted that, to maintain structural integrity, when APFM is removed, the absolute difference operation is adopted as a substitute. When AMG is removed, no attention mask is generated, and the auxiliary attention map loss is not computed. When AGRM is removed, the output of APFM is directly fed into the multiscale refinement fusion head.

As shown in Table VI, on the LEVIR-CD dataset, removing APFM reduces the $F1$ score to 90.50%, resulting in a noticeable performance degradation, which indicates that APFM plays a critical role in capturing shallow change features at the early stage. The performance drop is even more significant when AMG is removed, with the $F1$ score decreasing to 88.76%, representing the largest loss among all single-module ablations. This demonstrates that deep semantic attention maps are essential for globally guiding shallow feature learning. The absence of AMG prevents the model from establishing stable semantic consistency across different scales, leading to substantial increases in missed detections and false alarms. When AGRM is removed, the $F1$ score decreases to 90.33%,

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TABLE VII
RESULTS OF BACKBONE NETWORK ABLATION EXPERIMENTS USING 20% LABELED DATA ON THE LEVIR-CD DATASET

Backbone	F1	IoU	FLOPs(G)	Params(M)
Efficientnet-B0	91.48	84.31	9.39	16.57
Efficientnet-B1	91.25	83.90	11.43	26.59
Efficientnet-B2	91.64	84.57	12.38	31.34
Efficientnet-B3	91.60	84.51	15.71	43.32
Efficientnet-B4	91.61	84.53	21.54	70.73

TABLE VIII
RESULTS OF ABLATION EXPERIMENTS USING DIFFERENT THRESHOLD VALUES ON THE LVIR-CD DATASET WITH 10% LABELED DATA

Filter value	F1	P	R	IoU
NO	90.64	92.28	89.06	82.88
0.5	90.68	93.26	88.24	82.95
0.8	91.07	93.51	88.75	83.60
0.95	90.71	92.03	89.44	83.01
0.8-0.95	91.41	93.20	89.69	84.18

indicating that AGRM is crucial for fine-grained structural enhancement and boundary refinement. Without AGRM, the change saliency of shallow features is weakened, making it difficult for the model to detect small objects and subtle boundary variations.

On the WHU-CD dataset, the contributions of each module are similarly validated. When APFM and AMG are removed, the $F1$ scores decrease to 90.34% and 89.79%, respectively, indicating that cross-temporal feature modeling and semantic saliency guidance are also indispensable in complex urban scenes. Notably, when AGRM is removed, the $F1$ score still reaches 91.79%, suggesting that deep semantic constraints contribute more prominently to overall performance in the WHU-CD scenario. Nevertheless, the fine-grained refinement mechanism still plays a positive role in improving IoU.

When APFM, AMG, and AGRM are jointly employed, the model achieves the best performance. On LEVIR-CD, the $F1$ score increases to 91.41% with an IoU of 84.18%; on WHU-CD, the $F1$ score reaches 93.55% with an IoU of 87.88%. These results demonstrate that the three modules exhibit a strong synergistic and complementary relationship, collectively endowing AGNet with robust feature representation capability under the semi-supervised setting.

B. Backbone Network Ablation

To verify the impact of different backbone networks on AGNet's performance and computational complexity, ablation experiments were conducted on the LEVIR-CD dataset using 20% labeled data. Table VII lists evaluation metrics for EfficientNet variants from B0 to B4 as feature extractors.

The results show that as the network scales increase from B0 to B4, the number of parameters and FLOPs rise steadily, but the performance gain in $F1$ and IoU quickly saturates, with minor fluctuations in larger models. For example, EfficientNet-B0 achieves an $F1$ of 91.48%, whereas the larger EfficientNet-B4 achieves 91.61%, a difference of less than 0.2%. This indicates that the additional representation capacity of very large backbones does not translate into significant detection improvements but increases computational and storage costs. EfficientNet-B2 achieves the best tradeoff, with $F1$ of 91.64% and IoU of 84.57%, while maintaining only 12.38 G FLOPs and 31.34 M parameters. Larger models (B3 and B4) provide no substantial performance gain but increase computational burden, whereas the lighter B0 has the lowest complexity but slightly underperforms in IoU. Based on these observations, EfficientNet-B2 was selected as the standard backbone for AGNet.

C. Ablation of Pseudo-Label Filtering Thresholds

Table VIII presents the effect of different pseudo-label filtering thresholds on model performance using 10% labeled data on the LEVIR-CD dataset. Without any filtering strategy (NO), the model achieves $F1$ of 90.64%, indicating that using all pseudo-labels introduces noise and degrades performance.

Applying a fixed threshold of 0.8 significantly improves performance, raising $F1$ to 91.07%, showing that high-confidence pseudo-labels effectively reduce noise. Thresholds of 0.5 or 0.95 still outperform the no-filtering case, but either introduce low-quality labels or overly reduce the number of samples, leading to varying performance drops. When using a dynamic filtering range of 0.8–0.95, the model achieves the best performance, with $F1$ of 91.41% and IoU of 84.18%. This demonstrates that dynamic range filtering ensures sufficiently high-quality pseudo-labels, balancing stability and representativeness, and significantly enhances the robustness and generalization of the model in SSL.

VI. CONCLUSION

This article addresses several critical challenges commonly encountered in RSCD, including high annotation costs, severe pseudo-label noise, and the difficulty of modeling multiscale change features. To this end, we propose AG-SemiCD, an efficient change detection framework that integrates attention-guided feature aggregation with an adaptive SSL strategy.

In terms of model design, we constructed the AGNet. Through the synergistic action of the dual-stream APFM, the AMG, and the AGRM, the network achieves cross-scale interaction and refined fusion of deep and shallow semantics, significantly enhancing the identification capability for fine-grained change regions.

On the SSL side, we further introduce an A-EMA mechanism and a dynamic pseudo-label filtering strategy, which effectively suppresses pseudo-label noise accumulation, alleviates confirmation bias, and fully exploits latent information from unlabeled data. Extensive experiments on three public RSCD datasets (LEVIR-CD, WHU-CD, and MSRS-CD) demonstrate that AG-SemiCD achieves performance significantly superior to existing SOTA methods, particularly under low labeled ratio conditions, consistently ranking first across multiple key metrics. Notably, when using only 5%–10% labeled data, AG-SemiCD attains performance that is comparable to, or even matches, fully supervised counterparts, highlighting its exceptional data efficiency and strong generalization capability. These results indicate that AG-SemiCD provides a high-accuracy, low-annotation-cost, and

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deployment-friendly solution for practical RSCD applications. Looking ahead, we plan to further extend the application scope of the AG-SemiCD framework by generalizing it to change detection tasks involving heterogeneous optical [41] and SAR imagery [42].

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Shenbo Liu received the B.S. degree in electronic information science and technology from Changsha University of Science and Technology, Changsha, China, in 2021, where he is currently pursuing the Ph.D. degree in electronic science and technology.

His main research interests include computer vision, deep learning, and change detection.



Dongxue Zhao received the M.S. degree in electronic science and technology from Changsha University of Science and Technology, Changsha, China, in 2024, where she is currently pursuing the Ph.D. degree in electronic science and technology.

Her main research interests include computer vision, target detection, and heterogeneous accelerated neural networks.



Ying Tan received the B.S. degree in electronic information science and technology from Changsha University of Technology, Changsha, China, in 2023, where she is currently pursuing the M.S. degree in electronic science and technology.

Her main research interests include computer vision and change detection.



Lijun Tang (Member, IEEE) received the B.S. degree in physics from Xiangtan University, Xiangtan, China, in 1986, and the M.S. degree in nuclear physics and the Ph.D. degree in nuclear technology and application from Shanghai Institute of Atomic Nuclei, Chinese Academy of Sciences, Shanghai, China, in 1989 and 2003, respectively.

He is currently a Professor at Changsha University of Science and Technology, Changsha, China, and has presided over more than 20 vertical research projects, including the National Science and Technology Support Project, Hunan Provincial Natural Science Foundation Project, Department of Education Research Project, and Changsha Science and Technology Bureau Project, and has published more than 100 articles in domestic and international academic journals.

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